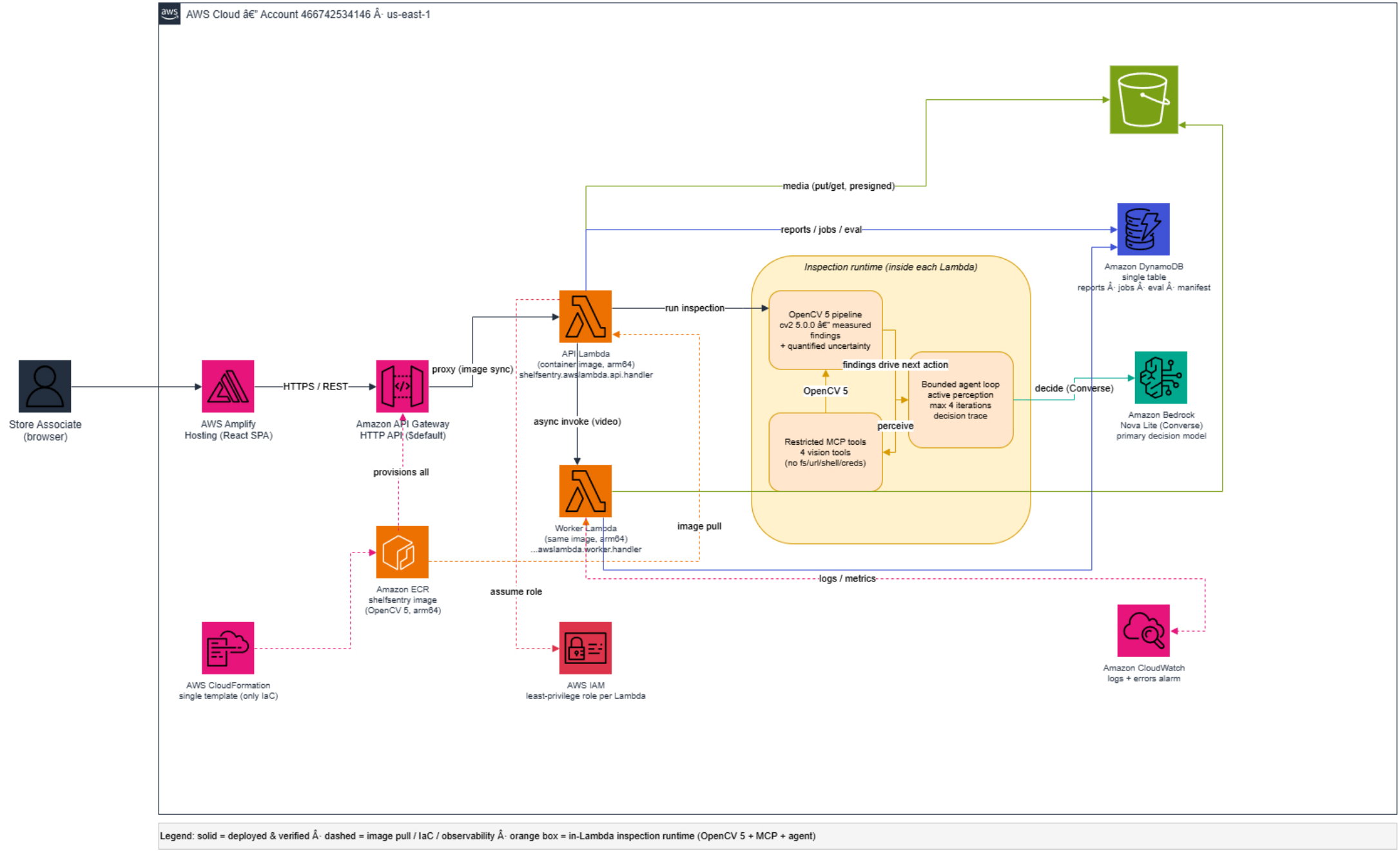
ShelfSentry AI - Active-Perception Shelf Inspection on AWS



Inspection

Upload a shelf image or video, or run the bundled sample, to start an active-perception inspection.

Shelf media

Image or video file

Choose File No file chosen

Accepted types: image/jpeg, image/png, image/webp, image/bmp, video/mp4, video/quicktime, video/x-msvideo. Maximum size 25 MB.

Start inspection | Run example (bundled sample)

No inspection yet

Choose a file or run the bundled sample to see results here.

Inspection

Upload a shelf image or video, or run the bundled sample, to start an active-perception inspection.

Shelf media

Image or video file

Choose File No file chosen

Accepted types: image/jpeg, image/png, image/webp, image/bmp, video/mp4, video/quicktime, video/x-msvideo. Maximum size 25 MB.

Start inspection | Run example (bundled sample)

âœ“ Inspection complete

Report id: `insp-20261025-cf9160807a7d`
Status: complete
Human review: Not required

Finalized after 4 decision(s) and 3 tool execution(s) (terminal action: finalize_report). Candidate regions are estimates with visual caveats; any empty-looking region is reported with its caveats and is not asserted as a proven out-of-stock condition.

View full report

Start another inspection

Inspection results

Recommendation

Flagged for human review: the agent found conflicting or insufficiently resolved evidence and is not asserting an out-of-stock condition. A reviewer should confirm.

▲ Needs human review — Human review required.

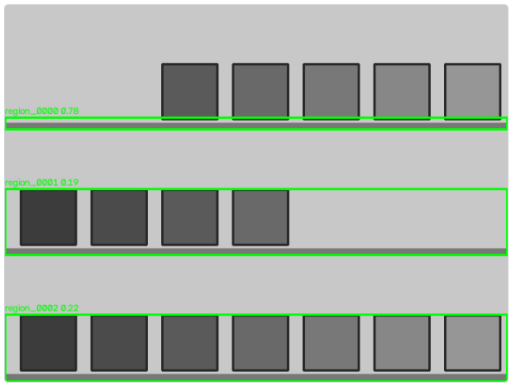
Visual evidence

Highlighted regions are classical computer-vision estimates, not certainties.

Original input

The original uploaded image is referenced by the report; the annotated output on the right marks each detected region.

Annotated output



Candidate regions

1. Region region_0000

Empty-space ratio	77.8% (estimate)
Uncertainty value	0.4282449417114258
Interpretation	No documented interpretation for this value.
Heuristic score	0.6544010416666667 (ordinal, not a probability)

Caveat: This region is an estimate of empty shelf space, not a confirmed out-of-stock condition. An empty-looking region may instead be shadow, occlusion, shelf dividers, or poor image quality.

2. Region region_0001

Empty-space ratio	18.9% (estimate)
Uncertainty value	0.5388661820123269
Interpretation	No documented interpretation for this value.
Heuristic score	0.3402144607843137 (ordinal, not a probability)

Caveat: This region is an estimate of empty shelf space, not a confirmed out-of-stock condition. An empty-looking region may instead be shadow, occlusion, shelf dividers, or poor image quality.

3. Region region_0002

Empty-space ratio	22.3% (estimate)
Uncertainty value	0.4458167379326663
Interpretation	No documented interpretation for this value.
Heuristic score	0.35391472868217055 (ordinal, not a probability)

Caveat: This region is an estimate of empty shelf space, not a confirmed out-of-stock condition. An empty-looking region may instead be shadow, occlusion, shelf dividers, or poor image quality.

Agent decision trace

Initial finding: `insp-20261026-cd8dd11c7a90-image`

Iteration 0 · agent decision

Action: `inspect_crop`

Rationale: Region with moderate uncertainty and heuristic score suggests further inspection is needed to confirm the condition.

Decided by bedrock

Iteration 0 · tool execution

Tool: `inspect_region_crop`

Result: ✓ ok (ref media://e4870687e94e79517d98d720af216680)

Iteration 1 · agent decision

Action: `request_human_review`

Rationale: The vision finding indicates high uncertainty in the estimated empty shelf space regions, and the confidence in the measurements is low. A human review is needed to resolve the ambiguity.

Decided by bedrock

History

Previous inspections with their status and time. Search and filter, then open any row to view its full results.

Search: Search recommendatioı

Status: All statuses

Show: All

Inspection	Status	Recommendation	Created
insp-20261026-cd8dd11c7a90	Needs review	Flagged for human review: the agent found conflicting or insufficiently resolved evidence and is not asserting an out-of-stock condition. A reviewer should confirm.	26/10/2026, 7:19:08 pm
insp-20261026-93021d0ba631	Needs review	Flagged for human review: the agent found conflicting or insufficiently resolved evidence and is not asserting an out-of-stock condition. A reviewer should confirm.	26/10/2026, 7:18:36 pm
insp-20261026-6bfbeac2cad6	Needs review	Flagged for human review: the agent found conflicting or insufficiently resolved evidence and is not asserting an out-of-stock condition. A reviewer should confirm.	26/10/2026, 6:50:41 pm
insp-20261026-b9b9bcd9f917	Needs review	Flagged for human review: the agent found conflicting or insufficiently resolved evidence and is not asserting an out-of-stock condition. A reviewer should confirm.	26/10/2026, 6:32:40 pm
insp-20261026-c9b21c2df8c0	Needs review	Flagged for human review: the agent found conflicting or insufficiently resolved evidence and is not asserting an out-of-stock condition. A reviewer should confirm.	26/10/2026, 5:46:03 pm
insp-20261026-7a0dfa9acada	Needs review	Flagged for review: the agent judged the input image too poor to assess reliably and requested a higher-quality image. No out-of-stock condition is asserted; re-capture and re-run.	26/10/2026, 5:40:06 pm
insp-20261026-eb27903ca7d7	Needs review	Flagged for human review: the agent found conflicting or insufficiently resolved evidence and is not asserting an out-of-stock condition. A reviewer should confirm.	26/10/2026, 5:39:00 pm
insp-20261026-1ff330069900	Needs review	Flagged for review: the agent judged the input image too poor to assess reliably and requested a higher-quality image. No out-of-stock condition is asserted; re-capture and re-run.	26/10/2026, 5:38:02 pm
insp-20261025-cf9160807a7d	Complete	Finalized after 4 decision(s) and 3 tool execution(s) (terminal action: finalize_report). Candidate regions are estimates with visual caveats; any empty-looking region is reported with its caveats and is not asserted as a proven out-of-stock condition.	26/10/2026, 12:29:26 am
insp-20261025-f8b159378eb8	Needs review	Flagged for human review: the agent found conflicting or insufficiently resolved evidence and is not asserting an out-of-stock condition. A reviewer should confirm.	26/10/2026, 12:23:59 am
insp-20261025-10074f5410ee	Needs review	Flagged for human review: the agent found conflicting or insufficiently resolved evidence and is not asserting an out-of-stock condition. A reviewer should confirm.	26/10/2026, 12:23:53 am
insp-20261025-7a9740530f13	Needs review	Flagged for human review: the agent found conflicting or insufficiently resolved evidence and is not asserting an out-of-stock condition. A reviewer should confirm.	26/10/2026, 12:19:28 am
insp-20261025-78199f25362c	Complete	Finalized after 2 decision(s) and 1 tool execution(s) (terminal action: finalize_report). Candidate regions are estimates with visual caveats; any empty-looking region is reported with its caveats and is not asserted as a proven out-of-stock condition.	26/10/2026, 12:18:34 am
insp-20261025-c079f3fbdd2e	Needs review	Flagged for human review: the agent found conflicting or insufficiently resolved evidence and is not asserting an out-of-stock condition. A reviewer should confirm.	25/10/2026, 11:52:24 pm
insp-20261025-0cee50567750	Needs review	Flagged for review: the agent judged the input image too poor to assess reliably and requested a higher-quality image. No out-of-stock condition is asserted; re-capture and re-run.	25/10/2026, 11:32:18 pm
insp-20261022-2fb9552567d0	Needs review	Flagged for review: the agent judged the input image too poor to assess reliably and requested a higher-quality image. No out-of-stock condition is asserted; re-capture and re-run.	23/10/2026, 4:05:20 am
insp-20261022-50893f972cfc	Complete	Finalized after 2 decision(s) and 1 tool execution(s) (terminal action: finalize_report). Candidate regions are estimates with visual caveats; any empty-looking region is reported with its caveats and is not asserted as a proven out-of-stock condition.	23/10/2026, 3:50:29 am
insp-20261022-14593223daf0	Needs review	Flagged for human review: the agent found conflicting or insufficiently resolved evidence and is not asserting an out-of-stock condition. A reviewer should confirm.	23/10/2026, 3:12:27 am
insp-20261022-4b9f5ef39713	Needs review	Flagged for human review: the agent found conflicting or insufficiently resolved evidence and is not asserting an out-of-stock condition. A reviewer should confirm.	23/10/2026, 3:10:11 am

Evaluation

Measured test-set performance from recorded evaluation runs. Every value shown is a measured result; no metrics are estimated or fabricated.

Run `eval-66bca98376a7`

○ DEMO DATA

8 labeled samples evaluated.

Measured performance summary

Metric	Measured value
Agent decision correctness	`100.0%`
False positives	`0`
False negatives	`2`
Processing latency (p50)	`68 ms`
Processing latency (p95)	`97 ms`

False-positive examples

No false-positive examples were recorded for this run.

False-negative examples

Sample	Expected	Actual	Note
`eval-gap-single-right`	has_gap=True (drew 1 gap(s))	has_gap=False	Stocked shelf with one empty gap on the middle-right row; a borderline case for the empty-ratio threshold, so its terminal action is excluded from the decision-correctness metric. The vision heuristic flagged no empty region although a gap was drawn (false negative).
`eval-gap-bottom`	has_gap=True (drew 1 gap(s))	has_gap=False	Stocked shelf with one wide empty gap on the bottom row; a borderline case for the empty-ratio threshold, so its terminal action is excluded from the decision-correctness metric. The vision heuristic flagged no empty region although a gap was drawn (false negative).

Documented failure cases and limitations

- The evaluation images are SYNTHETIC: they are drawn programmatically with NumPy and OpenCV, not photographed from real shelves. Their labels are the project authors' own ground truth for drawn content and are NOT real retail ground truth; measured performance here does not predict real-world accuracy.
- The detection metric scores a coarse per-image 'has_gap' boolean (does the vision pipeline flag any empty region at or above the configured empty-space-ratio threshold) against whether a gap was deliberately drawn. It is a heuristic sanity check, not a bounding-box-level accuracy measure, and it inherits the classical-CV heuristic's limitations.
- Agent decision correctness compares the active-perception loop's TERMINAL action against a hand-labeled expected terminal action, only for the subset of samples where a single expected terminal action is well defined. It measures whether the deterministic fallback policy routes each scene to the intended action; it does NOT measure the correctness of a learned/LLM policy and does not apply to samples left unlabeled.
- The set is deliberately small and reproducible (a handful of scenes) to stay fast and asset-free; it is not a statistically powered benchmark.
- Latency percentiles are wall-clock measurements taken on the machine that ran the evaluation (time.perf_counter around the real pipeline) and are therefore environment-dependent; absolute values will vary by hardware, load, and OpenCV build.
- Agent decision correctness was measured over 6 of 8 samples that carry an expected terminal-action label.

ShelfSentry AI

Input Image



ShelfSentry AI Result



Inspection Result

Overall Status

Possible Empty Region

Confidence

0.68 (heuristic)

Empty Space Ratio

0.72

Uncertainty

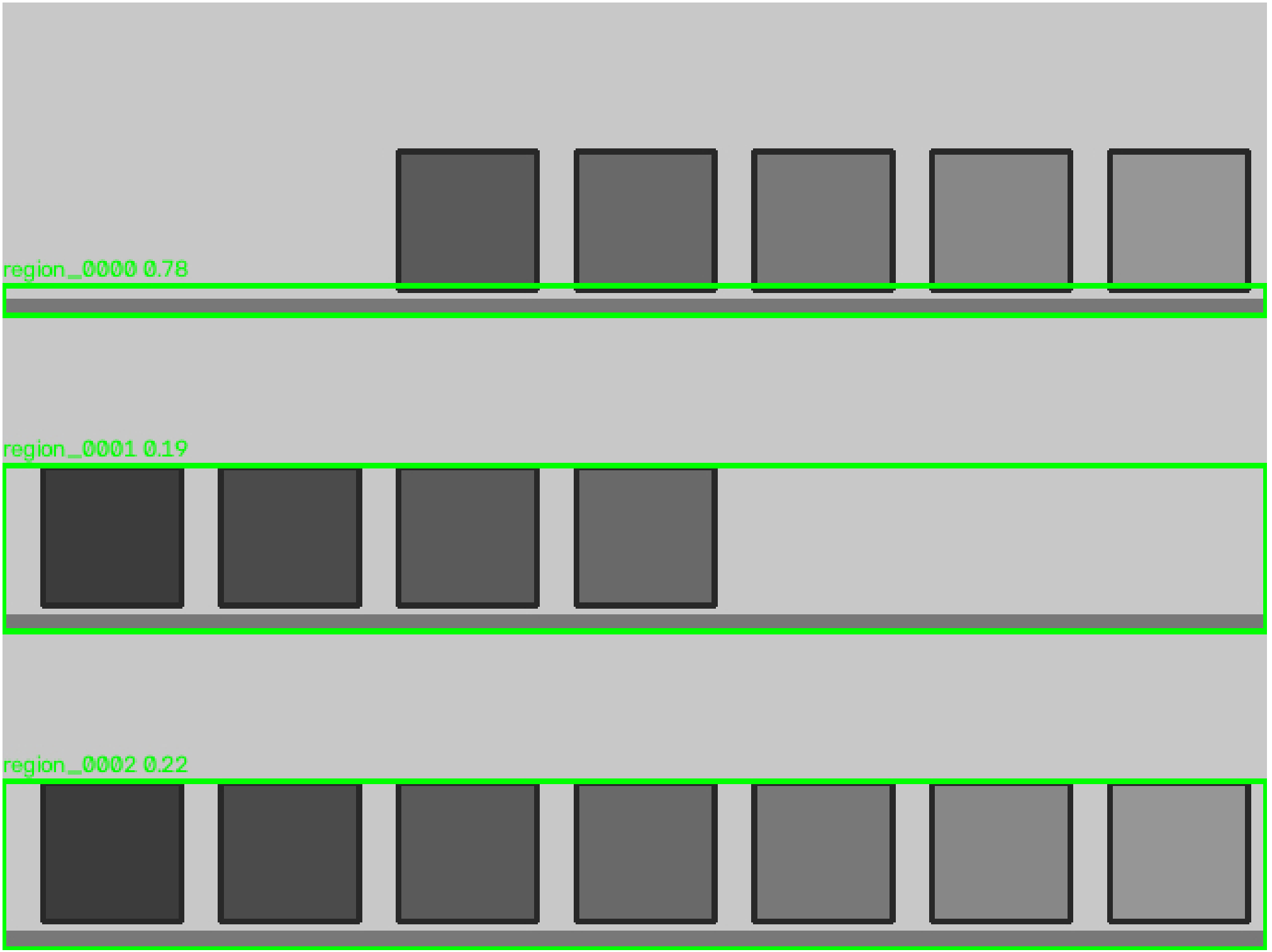
High

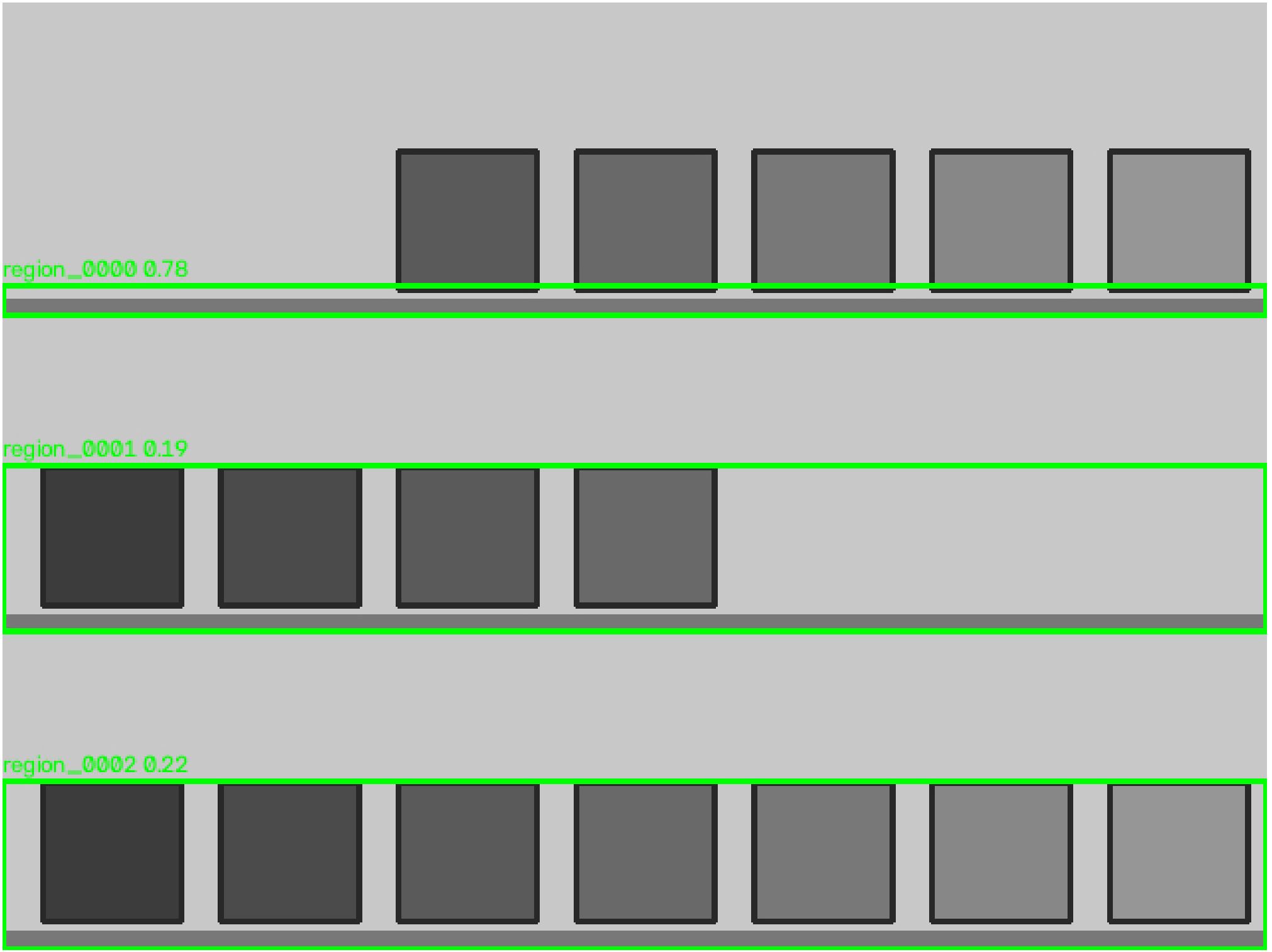
Recommendation

Inspect further

Decision Trace

1. **Detected candidate empty region**
   empty_space_ratio: 0.72, uncertainty: high
2. **Decided to inspect crop**
   decided_by: bedrock, action: inspect_crop
3. **Analyzed cropped region**
   tool: analyze_image, result: confirmed empty region
4. **Finalized report**
   decided_by: bedrock, action: finalize_report


region_0000 0.78
region_0001 0.19
region_0002 0.22


region_0000 0.78
region_0001 0.19
region_0002 0.22